OpenShift Disconnected Installation Guide

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Overview

Red Hat OpenShift is a comprehensive Kubernetes platform that simplifies the deployment, management, and scaling of applications, offering support for both containers and virtual machines within a single environment. In Internet-connected environments, OpenShift installation is straightforward with Installer-Provisioned Infrastructure (IPI), a guided, automated method ideal for cloud providers or on-premise setups. However, in disconnected environments, especially those with stringent security requirements, installations become more complex, requiring mirroring of all necessary content locally and tricks to simulate an internet connection for OpenShift's functionality.

Step 1 - Prerequisites for Disconnected Installation

Bastion Host Requirements (connected and disconnected)

- Operating system: RHEL 9
- 2 CPU 8 GB Memory (minimum recommendations)
- Storage minimum of 150 GB of available disk space for image content
- DNS resolvable forward/reverse (disconnected bastion only)
- Install podman and nmstate (disconnected bastion only)
- FIPS enabled if enabling FIPS for OpenShift cluster.
- Set umask to 0022
- Disable fapolicyd

None

```
sudo fips-mode-setup --check

sudo fips-mode-setup --enable

umask 0022

sudo systemctl disable fapolicyd

sudo dnf install podman nmstate
```

OpenShift Installation Requirements

- **Network** - for bare metal installations it is highly recommended for the ports that connect the bare metal servers to the backend switch to be trunk ports. This allows VMs inside OpenShift to attach to additional VLANs in much the same way as when using VMWare.
    - MAC address of each server will be needed at install time
    - DHCP or Static IP?
    - Bonded network interfaces? Active/Backup or LACP?
    - Identify VLANs?
    - Will a proxy be used, transparent or non-transparent?
    - MTU?
- **NTP** - OpenShift nodes require strict clock synchronization for security handshake protocols and etcd consensus. NTP can be configured through the DHCP Options or added to the agent-config.yaml at install time.
- **DNS**- Each OpenShift host must have a unique IP/DNS resolution
    - DNS forward/reverse entries required for the following entries:
        - api.<cluster_name>.<base_domain>
        - api-int.<cluster_name>.<base_domain>
        - *.apps.<cluster_name>.<base_domain>
        - hostname for each bare metal server
- **Storage -** OpenShift root volumes or ODF (optional) cannot be spinning disks. They must be either SSD or NVME. Recommend RAID 1 for root volume with bare metal installations.
    - Obtain root volume /dev/disk/by-path/<device_name>
        - Example: if installing to /dev/sdd run the following command on the host: ls -l /dev/disk/by-path/ | grep sdd

Step 2 - Download and Configure Software on the Connected Bastion Host

1. Download OpenShift Software

```
None
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/op
enshift-client-linux-amd64-rhel9.tar.gz
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/oc
-mirror.rhel9.tar.gz
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/op
enshift-install-rhel9-amd64.tar.gz
$ wget
https://developers.redhat.com/content-gateway/file/pub/cgw/mirror-registry/2.0.
10/mirror-registry-amd64.tar.gz
```

2. Acquire Pull Secrets

We will need your pull secret to authenticate with the Red Hat servers. In the same folder issue the following commands:

```
None
$ mkdir ~/.docker
$ echo '<YOUR_RED_HAT_PULL_SECRET>' > ~/.docker/config.json
```

3. Configure oc-mirror and download Kubernetes Operators

Red Hat created a tool called oc-mirror to help package up all needed components of openshift. Oc-mirror uses a config file, called imageset.yaml. This config file tells the tool exactly what components to pull down.

The following imageset config file is an example to pull the containers and operators. If you need additional operators that are not included in this file we will explain how to accomplish this in the following section.

```
None
#This will be the imageconfig.yaml. Cut-and-paste into your command line.

$ cat >> imageset.yaml<< EOF
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
archiveSize: 4 # Archive chunk size in GB
mirror:
  platform:
    architectures:
      - amd64
    channels:
    - name: stable-4.21
      type: ocp
      minVersion: 4.21.5
      maxVersion: 4.21.5
    graph: true
  operators:
  - catalog: registry.redhat.io/redhat/redhat-operator-index:v4.21
    packages:
      - name: cincinnati-operator
      - name: compliance-operator
      - name: web-terminal
      - name: local-storage-operator
  additionalImages:
  - name: registry.redhat.io/ubi8/ubi:latest
  - name: registry.redhat.io/ubi9/ubi:latest
  - name: registry.redhat.io/rhel8/support-tools
  - name: registry.redhat.io/rhel9/support-tools
EOF
```

To list all operators available in all catalogs, use the following command which will display all operator names for OpenShift 4.21 and save them as a text file in your local directory. Update your imageconfig.yaml with the appropriate operators you want.

None

```
$ for i in $(oc-mirror list operators --catalogs --version=4.21 | grep
registry); do $(oc-mirror list operators --catalog=$i --version=4.21 > $(echo
$i | cut -b 27- | rev | cut -b 7- | rev).txt); done
```

4. Run oc-mirror command - Mirror to Disk

On the connected bastion we will run the oc-mirror command, and pull down all images that are needed to stand up OpenShift + Day 2 operations. This process may take a while. We also need to ensure that umask is set to 0022 on STIG'd RHEL servers to allow the oc-mirror process.

None

```
$ umask 0022
$ oc-mirror --v2 --authfile /home/ec2-user/pull-secret.json --config
./imageset-config.yaml --cache-dir /home/ec2-user/.oc-mirror/.cache
file:///home/ec2-user/mirror/
```

Step 3 - Transfer Software to Disconnected Network

From here we need to transfer the files over to the disconnected side however you see fit - rsync, scp, sneakernet, burn to a DVD, host it in an apache server, etc. 👟

We will need to move over the entire directory to include the oc, oc-mirror, mini-quay, etc. binaries, along with the imageset-config and mirror_seq tarballs. Below is the scp command to transfer files over from the connected STIG'ed bastion to the disconnected STIG bastion.

None

```
# scp -r * <username>@<disconnected_bastion_host>:.

$ scp -r * ec2-user@airgap-stig:.
```

Step 4 - Configure Disconnected Bastion Host

On the disconnected side let's unzip our ingredients and check to make sure everything is good to go.

```
None
#Untar our ingredients
$ tar xvf mirror-registry.tar.gz
$ tar xvf oc-mirror.tar.gz
$ tar xvf openshift-client-linux.tar.gz
$ tar xvf openshift-install-linux.tar.gz

#Move the tools and change ownership make executable
$ sudo cp {mirror-registry,oc,oc-mirror,openshift-install} /usr/local/bin/
$ sudo chown -R $USER
/usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}
$ sudo chmod +x /usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}
$ sudo restorecon -v
/usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}

#We need to disable FApolicyD for the install
$ sudo systemctl stop fapolicyd.service

#The STIG sets max_user_namespaces to 0, preventing rootless containers. Do the
following to enable rootless containers

#Query the current value
$ sysctl -a | grep max_user_namespaces
user.max_user_namespaces = 62372

#Query / change the stored value
$ grep namespaces /etc/sysctl.d/99-sysctl.conf

#Apply the new value
$ sysctl -p /etc/sysctl.d/99-sysctl.conf

# Update /etc/subuid and /etc/subgid with the rootless podman user's id info
```

Step 5 - Installing the Mirror Registry on Bastion Host

As OpenShift is deployed as a set of containers, a registry is necessary to operate properly. In connected environments, OpenShift would use Red Hat's container registry for the initial installation but in disconnected environments this will not be available. Therefore, we need to stand one up to serve this purpose. The installation of the mirror-registry should be done using a general user account

The STIG modifies the user bashrc and profile to default to 0077. During the mirroring process to your local registry, we also build your default catalog source. During that process, we need to ensure that the umask is set to 0022 so that OpenShift can read those files within the built container. This is necessary because, by default, OpenShift cannot run containers as root for security reasons.

1. Install Quay

```
None
#We had to change the umask from 0077 to 0022 for the quay install, due to the
fact that it always ssh'es into localhost and it get the umask from the default
/etc/bashrc and /etc/profile
sudo sed :%s/077/022/gc /etc/bashrc
suod sed :%s/077/022/gc /etc/profile

#Install the mirror registry (Quay)
#We will need to open up port 8443 for Quay
sudo firewall-cmd --add-port 8443/tcp --permanent
sudo firewall-cmd --reload

#If you have a cert and key that you would like to use instead of having Quay
create a self-signed one please add the following
---
      --sslCert string          The path to the SSL certificate Quay should use
      --sslCheckSkip            Whether or not to check the certificate
hostname against the SERVER_HOSTNAME in config.yaml.
      --sslKey string           The path to the SSL key Quay should use
  -H, --targetHostname string   The hostname of the target you wish to install
Quay to. This defaults to $HOST (default
"rhel8-mirror-stig.airgap.dota-lab.iad.redhat.com")
  -u, --targetUsername string   The user on the target host which will be used
for SSH. This defaults to $USER (default "allen")
----
```

```
mkdir -p /data/mirror-registry
./mirror-registry install --quayHostname <Full FQDN> --quayRoot
/data/mirror-registry --quayStorage /data/mirror-registry --initUser admin
--initPassword redhat123
```

2. Test if Quay is up and functioning

```None
#Copy certificates that created - Only do this if Quay created the certs for
you. Quay will create self-signed certs during the install process.

sudo cp -v /data/mirror-registry/quay-rootCA/rootCA.pem
/etc/pki/ca-trust/source/anchors/
sudo update-ca-trust
podman login <Full FQDN>:8443
```

3. Upload the Bundled Images into the Registry

```None
#Starting the intital image transfer into local Quay - this may take a while
oc-mirror -c imageset-config.yaml --from file:///<path-to-mirror>/
docker://<Full FQDN>:8443 --v2

#You should notice that your local Quay would start showing images under
https://<Full FQDN>:8443/repository

# Once you recieve message such as  images mirrored successfully followed by
Goodbye, thank you for using oc-mirror the disk-to-mirror is complete
```

Step 6 - Create install-config

Let's create the install-config; the install config is the file that tells CoreOS how to configure itself upon installation. You can find sample install-config.yaml HERE

1. Grab Pull Secret

First we need to grab the pull secret that will be copied into the config.

```
None
#We will need to get our auth from our local quay registry
$ podman login --authfile config.json <Full FQDN>:8443
$ jq -c -r '.' config.json ###This is your $<PULL_SECRET> that you will need to
copy
```

2. SSH keypair is needed for access to CoreOS

You will need to provide the contents of the .pub key during the install-config create process

```
None
ssh-keygen

Generating public/private rsa key pair.
Enter file in which to save the key (/home/user/.ssh/id_rsa):
/home/user/.ssh/id_rsa_openshift
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/user/.ssh/id_rsa_openshift
Your public key has been saved in /home/user/.ssh/id_rsa_openshift.pub    |
```

3. VMware account with required permissions

The openshift-installer will connect to vSphere during the install-config process to identify resources such as storage and network. Here is a list of permissions required

4. Run openshift-installer to generate install-config.yaml

The command below will walk you through a series of steps for required installation configurations

```
None
mkdir openshift-configs

openshift-install-fips create install-config
? SSH Public Key /home/mikehall/.ssh/id_rsa_openshift.pub
? Platform vsphere
? vCenter [? for help]
```

4. Add Registry CA and ImageDigestMirrors to install-config.yaml

For disconnected installations we need to tell the openshift-installer where to get the container content and provide the CA for the registry. The ImageDigestMirrors release configuration is found in the working-dir/cluster-resourcs/idms.yaml file where the mirror sequence was imported from. Add the following entries to the bottom of the install-config.yaml file.

```
None
imageDigestMirrors:
  - mirrors:
    - mirror-registry:8443/openshift/release
    source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
  - mirrors:
    - mirror-registry:8443/openshift/release-images
    source: quay.io/openshift-release-dev/ocp-release
additionalTrustBundle: |
  -----BEGIN CERTIFICATE-----
  -----END CERTIFICATE-----
```

Step 7 - Install the cluster

Create a directory called cluster and copy the install-config.yaml into it. The openshift-install will consume the install-config.yaml during the create cluster process.

```None
mkdir cluster

cp install-config.yaml cluster

openshift-install-fips create cluster --dir ./cluster --log-level info
```

Post Installation Tasks

1. Disabling the default sources

Operator catalogs that source content provided by Red Hat and community projects are configured for OperatorHub by default during an OpenShift Container Platform installation. In a restricted network environment, you must disable the default catalogs as a cluster administrator. You can then configure OperatorHub to use local catalog sources. Additionally we will want to remove the default helmchart and devfile repositories.

```
None
#Disable the sources for the default catalogs by adding disableAllDefaultSources:
true to the OperatorHub object:
export KUBECONFIG=auth/kubeconfig

oc patch OperatorHub cluster --type json \
    -p '[{"op": "add", "path": "/spec/disableAllDefaultSources", "value": true}]'

oc patch helmchartrepository openshift-helm-charts --type='merge' -p
'{"spec":{"disabled":true}}'

oc patch console.operator.openshift.io cluster --type='merge' -p
'{"spec":{"plugins":["dev-console-managed-devfile-catalog-disabled"]}}'
```

2. Updating the Catalog Source

We will need to tell OpenShift that we have a local mirror of all of the operators. Within your bastion host, you could see that oc-mirror created a directory called working-dir/cluster-resources. We will need to apply that directory.

```
None
$ oc apply -f <path-to-mirror>/<path-to-mirror>/working-dir/cluster-rsources
```

Troubleshooting

Troubleshooting with common problems when doing bare metal OCP installs.

- DNS issues (make sure your bastion host uses the same DNS as the servers)
- Certs do not link up with NTP, cert is ahead of NTP
    - If needed add NTP ip to agent-config.yaml
- The registry is not up
    - Certs
    - Filesystem issues with STIG
    - Ownership of files
- The pull-secret is not updated with the correct authentication JSON
- Agent-config.yaml with interfaces not called out in both "the upper section and lower section"
- MAC Addresses on agent-config.yaml are correct

References and Documentation

Red Hat Disconnected docs:
https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/html/disconnected_environments/about-disconnected-environments
(Or replace with your OCP Version)
https://www.redhat.com/en/blog/red-hat-openshift-disconnected-installations

Red Hat Openshift Docs
https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/

Red Hat Offline Knowledge Portal
https://docs.redhat.com/en/documentation/red_hat_offline_knowledge_portal/1